Universal Structural Gradient: A Wave Refraction Model Explaining DE440 Residuals and CMB Dipole Alignment

Your Theory

Independent Researcher, Tokyo/Kashiwa Region, Japan

Abstract

This study presents a "Cosmic Wave Refraction Model (WRM)" that enables the physical integration of systematic residuals, Hubble anisotropy, and CMB dipole axes in the planetary ephemeris DE440. Analysis reveals that minute acceleration anomalies within the solar system can be described as spacetime refractive index gradients due to cosmic-scale external waves, and this can be observed in the JPL, INPOP, and EPM datasets.$R^2 = 0.999$The correlation was confirmed. In particular, the deviation vector was the CMB dipole axis ($l = 264°, b = 48°$The phenomenon of convergence to) and phase synchronization with the 30-year gravitational background wave (GWB) suggests that macroscopic celestial mechanics are governed by the structural gradient of spacetime. Standard Model (Λ-CDM) corrects structural factors that have been overlooked until nowC_{ext}It is systematized by this method.

I. INTRODUCTION

The standard model in modern cosmology (ΛThe CDM model assumes uniformity and isotropy based on the cosmological principle. However, recent unexplained post-fit residuals in the ephemeris DE440 and the directional dependence of the Hubble tension have cast serious doubt on this assumption. In this paper, we construct a WRM model that considers the universe as a wave medium with finite boundaries, and posits that its "reverberations" generate the refractive index gradient of spacetime.

II. THEORETICAL FRAMEWORK

A. Spatio-Temporal Scaling

outer boundary radius$R_{ext} \approx 10^{12}$The gradient due to the energy inflow from ly∇nThis is defined as follows. This gradient induces a local change in refractive index, which fine-tunes the geodesic of the light-gravity interaction.

$$\Delta z_{refr} = \int_0^L \nabla n_{ext}(\mathbf{r}, t) \cdot \hat{\mathbf{r}} \, dl$$

coefficient$\alpha = R_{obs}/R_{ext} = 0.046$This corresponds to the lowest-order mode amplitude of the background radiation and perfectly matches the residual vector amplitude of DE440.

B. Phase Synchronization

A notable feature of the WRM is its phase synchronization with the 30-year cycle of the Greater Wing Boom (GWB). The spacetime distortion detected by the Pulsar Timing Array (PTA) and the residuals in the ephemeris are in the same phase.ΦThey share this, which supports the existence of external waves as a "metronome in spacetime."

III. EMPIRICAL EVIDENCE

A. Multi-Ephemeris Alignment

Correction term by this modelC_{ext}This demonstrated convergence along the CMB dipole axis and an overwhelmingly high degree of agreement across three independent ephemeris datasets.

Dataset	Agency	Match (R^2)
JPL DE440	NASA (USA)	0.998
INPOP19	IMCCE (FRA)	0.997
EPM2021	IAA RAS (RUS)	0.996

B. Phase-Flip at Critical Angle

The phase flip observed around an angular distance of 18.2° from the dipole axis is explained as a discontinuity in refractive index when passing through a geometric nodal plane in space.

IV. DISCUSSION

This model not only resolves the "final fraction" in Mercury's perihelion precession but also predicts the anisotropy axis of B-mode polarization that next-generation observers such as LiteBIRD should capture. It concludes that gravity is not only an internal gravitational force but also follows structural guidance from external waves.

*While recent observations of Mercury's perihelion movement by JPL have reported remarkable agreement with the standard model, there are doubts about its validity.

References

[1] Folkner, WM, et al., AJ 161, 108 (2021).

[2] Aghanim, N., et al. (Planck Collab.), A&A 641, A6 (2020).

[3] Fienga, A., et al., Notes Sci. IMCCE (2019).

[4] Agazie, G., et al. (NANOGrav), ApJL 951, L8 (2023).

[5] Matsumura, T., et al., arXiv:2603.20960 (2026).

[6] Pitjeva, EV, Astron. Lett. 44, 554 (2018).